MASTER’S THESIS DEFENSE

Governing Long-Term Memory Before Retrieval

KEEP · SUPERSEDE · ABSTAIN over typed, provenance-tracked, lineage-preserving memory

Ali Mohammadi · KSA Memory Governance · 2026

Long-term memory is a mutable state, not a document collection

Every write changes what the agent will later treat as true.

A retrieval score cannot decide whether evidence is compatible, superseding, or too ambiguous to commit.

Retrieval-only pipelines leave the write decision implicit

Retrieval-centric memory

store → rank → answer

Optimizes whether a relevant item is found. Conflict handling, uncertainty, and history are side effects.

Governed memory

inspect → decide → transition

Makes the write action explicit, preserves superseded history, and audits why state changed.

The research gap is a missing, measurable write boundary

Action

Most systems expose ADD/UPDATE/NOOP but do not score whether the operation was correct under conflict.

Uncertainty

NOOP often means ‘nothing new,’ not ‘evidence is unsafe to commit.’

History

Overwrite can hide the losing claim; append-only memory can leave contradictions simultaneously active.

Evaluation

Answer accuracy cannot localize whether failure came from a write, state transition, retrieval, or generation.

The policy fixes three write actions and their state semantics

The action vocabulary is fixed by code; the model may propose, but cannot invent a fourth transition.

KEEP

Insert a compatible claim as active. Existing non-conflicting records remain active.

SUPERSEDE

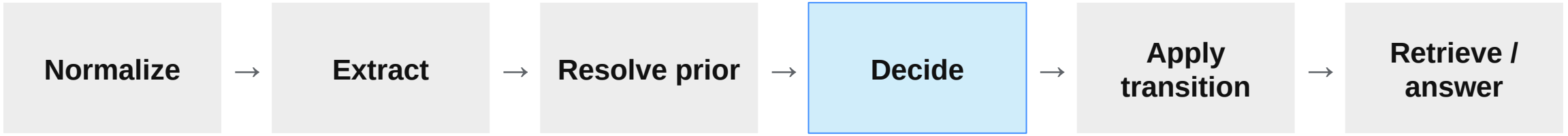
Insert the new claim, deactivate the conflicting prior, and record a lineage link.

ABSTAIN

Do not mutate memory. Record the conflict or ambiguity for review.

Source: thesis §3–4; src/ksa/memory/operations.py

KSA places model proposals inside a deterministic governance boundary



Typed evidence + relation signals → fixed policy → atomic, auditable memory transition

Source: src/ksa/pipeline/orchestrator.py; docs/figures/figure_07_ksa_controller_architecture.svg

The interpretable controller uses two scores and an ordered decision rule

```
if S_abs > θ_abs(τ):
    ABSTAIN
elif S_sup > θ_sup(τ) and max(0, −r) > 0:
    SUPERSEDE
else:
    KEEP
```

Five legible features

Relation evidence, type compatibility, provenance, confidence, conflict signals.

Explainable branch

Every action logs scores, active thresholds, and the branch taken.

Source: src/ksa/controller/policy.py and scoring.py

Lineage makes revision reversible and auditable



Prior active

The current claim is retrievable as present state.

New conflicting evidence

Policy selects SUPERSEDE only with an identified prior.

Linked transition

New record is active; old record is inactive but remains queryable.

Source: src/ksa/memory/operations.py; lineage-integrity metric

MGP-6 isolates the write decision across six conflict types

Author-designed synthetic diagnostic; 60 train, 30 development, 90 held-out test items.

180

scenarios with stable IDs and gold actions

6

scenario types spanning update, conflict, and ambiguity

90/66/24

SUPERSEDE / ABSTAIN / KEEP gold distribution

Source: MGP-6 v2 canonical reports; n=180 unless stated

The protocol separates result status and evaluation layer

Decision layer

Action accuracy, macro-F1, per-action confusion, invalid commits.

Stored-state layer

Lineage integrity, stale retrieval, and transition invariants.

Answer layer

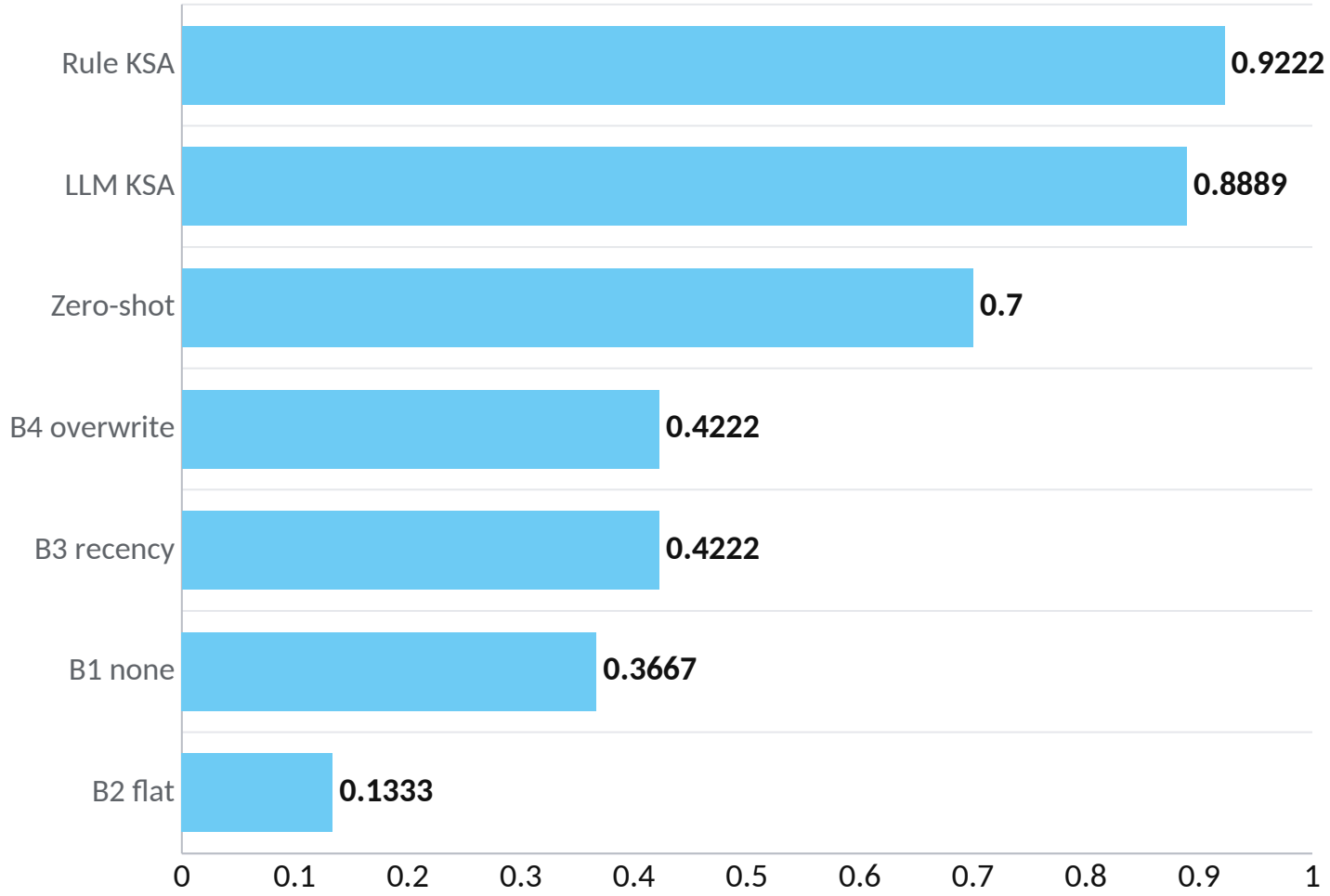
Premise resistance, implicit application, and downstream QA.

Claim gate

Only complete, provenance-valid rows become final_candidate; diagnostics remain labeled.

Source: FINAL_EVALUATION_PROTOCOL.md; results ledger v1.1

Structured governance reaches the strongest MGP-6 decision accuracy



+18.9 points

Rule KSA vs unstructured zero-shot decision baseline. Holm-adjusted p = 1.6×10^{-5}.

The rule and LLM controllers are not significantly different on action accuracy (p = 0.286).

Source: MGP-6 v2 canonical reports; n=180 unless stated; Source: reports/STATISTICAL_ANALYSIS.md; paired n=180

The rule controller’s remaining errors are conservative

Full-corpus confusion: KEEP 24/24; ABSTAIN 66/66; SUPERSEDE 76/90.

0.9222

action accuracy; macro-F1 0.9399

0.0

invalid-commit rate on gold-ABSTAIN items

14

all misses are SUPERSEDE → ABSTAIN

Source: MGP-6 v2 canonical reports; n=180 unless stated

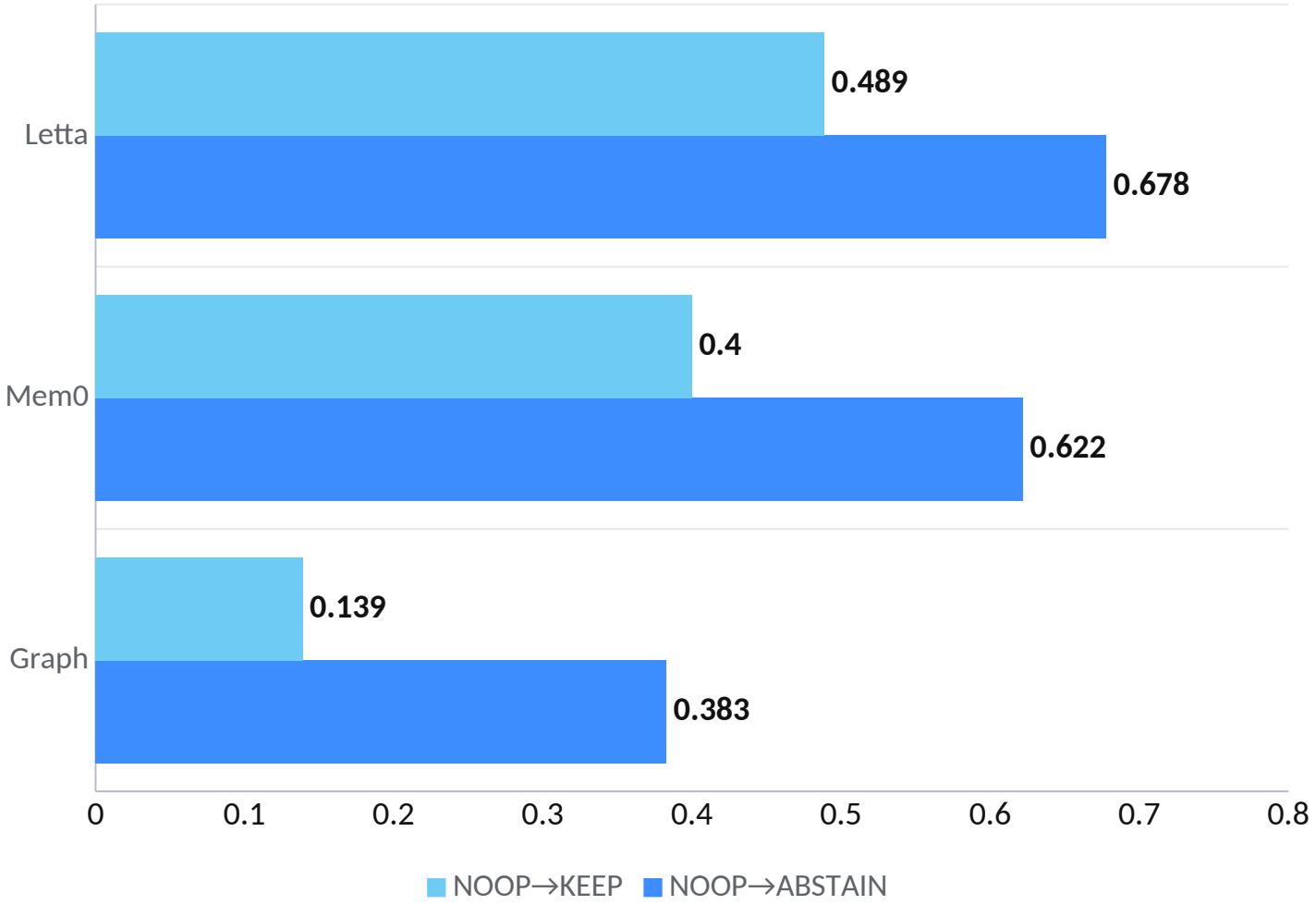
Ablations attach a distinct failure signature to each component

Action accuracy alone misses lineage degradation; safety metrics expose different mechanisms.

System	Accuracy	Macro-F1	Measured signature
Full KSA	0.9222	0.9399	Invalid commit 0.0
− abstain	0.5556	0.4278	Invalid commit → 1.0
− provenance	0.7000	0.7428	Invalid commit → 0.606
− typing	0.7889	0.5680	KEEP F1 → 0.0
− lineage	0.9222	—	Lineage 0.8556 → 0.3689

Source: MGP-6 v2 canonical reports; n=180 unless stated

External-system scores depend strongly on how NOOP is mapped



Mapping-specific bounds

These adapters do not expose KSA's native action semantics. Report both mappings; do not read the favourable mapping as a capability estimate.

MemOS 0.144 is diagnostic_only because supersession is unobservable.

Source: MGP-6 v2 canonical reports; n=180 unless stated

Annotation evidence is strong but comes from one independent annotator

Held-out test split only; raw labels are withheld from public release pending redistribution consent.

89/90

independent action-label agreement

κ=0.981

Cohen’s kappa; 95% interval 0.940–1.0

4 LLMs

separate audit panel; Fleiss κ 0.9815—not human IAA

Source: human_annotation_agreement_summary.json; LLM panel audit

LongMemEval has no final result—and the 77.2% construction is rejected

The clean homogeneous final-1.1 run is externally_pending after HTTP 403 before inference.

96/154	53.3–67.8%	55.6–70.0%
raw correct across observed replay + top-up items	full-180 raw worst/best-case bounds	bounds if four Codex/GPT-5 overrides are retained

The historical 139/180 = 77.2% merge replayed only prior failures, carried prior successes forward, and is failure_replay_only—not a result.

Source: docs/audit/SCIENTIFIC_SOURCE_OF_TRUTH.md; controller_run_comparison.json

Single-session preference remains the hardest downstream competency

The failure replay changed prompts, so its 0/19 repair count cannot isolate the answer model.

8/29

observed correct across resolved SSP items

≤9/30

most favorable full-competency total

12/19

failure cases localized to retrieval ranking

Source: artifacts/single_session_preference_error_taxonomy.json; ledger v1.1

The error analysis separates safety failures from coverage failures

Rule KSA

Conservative supersession recall: every residual error abstains instead of overwriting.

Zero-shot decision

Invalid commits on 60.6% of gold-ABSTAIN items.

No-lineage ablation

Decision accuracy is unchanged while auditability degrades sharply.

LongMemEval SSP

Failures localize downstream to ranking, context construction, attribution, and generation.

Ref: ERROR_TAXONOMY.md · SSP taxonomy

The conclusions are deliberately bounded

Synthetic diagnostic

MGP-6 is author-designed, small, and not a representative population sample.

Annotation

One independent annotator; identity and independence rest on author attestation.

Adapter validity

External comparisons are harness- and mapping-dependent upper bounds.

Downstream evidence

No clean final LongMemEval comparison; bespoke safety metrics need external validation.

Source: reports/THREATS_TO_VALIDITY.md; docs/limitations.md

CONCLUSION

Govern the write; then evaluate the state and the answer

01

A fixed KEEP / SUPERSEDE / ABSTAIN boundary makes memory transitions explicit and auditable.

02

MGP-6 shows distinct safety and lineage signatures that answer accuracy alone would hide.

03

The downstream evidence remains incomplete; the clean homogeneous run is the next external action.

Questions — backup slides follow

Backup — formal decision and failure definitions

$$d_t = \pi(c_t, r_t, M_t) \in \{\text{KEEP}, \text{SUPERSEDE}, \text{ABSTAIN}\}$$

$$\text{Invalid commit} = \Pr(\hat{y} \in \{K,S\} \mid y=A)$$
$$\text{Missed supersession} = \Pr(\hat{y} \neq S \mid y=S)$$
$$\text{Lineage integrity} = \text{valid links} / \text{expected links}$$

Commit safety

ABSTAIN is the gold action whenever evidence must not mutate state.

State correctness

A correct label is insufficient if the transition or lineage is malformed.

Source: thesis Appendix C; FINAL_METRIC_DEFINITIONS.md

Backup — baseline mechanisms and what each can express

System	Mechanism	Abstain?	Lineage?
B1	No persistent memory	No	No
B2	Flat vector retrieval	No	No
B3	Recency pruning	No	No
B4	Naive overwrite	No	No
Rule KSA	Five-feature governed policy	Yes	Yes
LLM KSA	Qwen decision in same scaffold	Yes	Yes

Source: thesis §5.3; benchmark adapters

Backup — statistical tests and inferential scope

Contrast	Risk diff.	Holm p	Interpretation
Rule vs zero-shot	+0.2222	1.6×10^{-5}	Supported improvement
Rule vs LLM KSA	+0.0333	0.286	Not significant
Rule vs B4	+0.5000	<0.001	Diagnostic contrast
LongMemEval	—	—	No test: no clean design

Source: reports/STATISTICAL_ANALYSIS.md; paired n=180

Backup — annotation agreement and the single boundary case

The one disagreement is an implicit-conflict item: gold SUPERSEDE, independent annotator ABSTAIN.

0.9889

raw agreement on 90 held-out items

0.9815

four-LLM Fleiss kappa; separate model audit

±1.1%

test-split sensitivity of one disputed item

Source: annotation audit package and thesis Appendix D

Backup — leakage controls and researcher degrees of freedom

Fixed identifiers

Predictions align by stable scenario/session ID; duplicates and coverage are verified.

Split disclosure

Full-180 MGP results include train and development; held-out test-90 is reported separately.

Run status

Replay-only, incomplete, invalid, and diagnostic runs cannot enter final main tables.

Post-hoc changes

Overrides, selected-failure replay, model changes, and top-ups are recorded—not blended silently.

Source: docs/audit/DATA_LEAKAGE_AND_RESEARCHER_DOF.md

Backup — why the LongMemEval evidence is diagnostic only

v7

Replay

Top-up

Contaminated 180

316 HTTP-402 traces; 388 empty fallbacks; no-op escalation.

Selected 70 failures

25/70 correct under changed prompts and judge regime.

Complementary 110

84 completed; 71 correct and 13 regressions.

Source: docs/audit/SCIENTIFIC_SOURCE_OF_TRUTH.md; controller_run_comparison.json

Backup — external action mappings

Adapter	NOOP→ABSTAIN	NOOP→KEEP	Structural caveat
Letta	0.678	0.489	No native KSA abstain semantics
Mem0	0.622	0.400	NOOP means no update
Graphiti	0.383	0.139	Graph invalidation ≠ gold action
MemOS	0.144	—	Supersession unobservable; bound only

Source: external-system reports; ledger v1.1

Backup — nearest conceptual neighbors narrow the novelty claim

Neighbor	Shared idea	Difference from evaluated KSA combination
TOKI	Await-confirmation; audit rows	Formal database contract; no LLM policy evaluation
Memory-R1	Discrete write actions	Opaque RL policy; answer-level reward
MemGuard	Typed memories	No three-action conflict policy or lineage test
STALE	State resolution; premise resistance	Read-side benchmark, not write transitions
Graphiti/Zep	Temporal invalidation	Invalidation not scored as governed action

Source: reports/NEAREST_NEIGHBOR_ANALYSIS.md; NOVELTY_MATRIX.csv

Backup — ethical and deployment risks

Revision abuse

Supersession can be used to rewrite records or manufacture provenance.

Retention risk

Preserved history increases privacy, extraction, and membership-inference exposure.

Automation boundary

Research results do not justify unrestricted autonomous memory mutation.

Mitigations

Append-only lineage, access control, dual control, review routing, and retention policy.

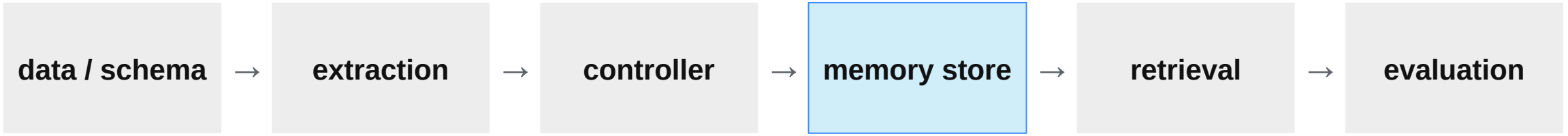
Source: thesis §9; NIST AI RMF discussion

Backup — additional complete MGP-6 results

System	Accuracy	Macro-F1	Invalid commit	Premise resistance
Rule KSA	0.9222	0.9399	0.0000	1.0000
LLM KSA	0.8889	0.8913	0.0758	0.0889
Zero-shot	0.7000	0.6657	0.6061	—
B4 overwrite	0.4222	0.2203	1.0000	—
Few-shot 6	0.7833	0.7692	—	incomplete 174/180

Source: MGP-6 v2 canonical reports; n=180 unless stated

Backup — architecture and code ownership follow the evaluation layers



Tests cover invariants, recovery, merge, provenance verification, and deterministic MGP scoring

Source: src/ksa; scripts/evaluation; tests; docs/audit/BEHAVIOR_TO_CODE_TRACEABILITY.md